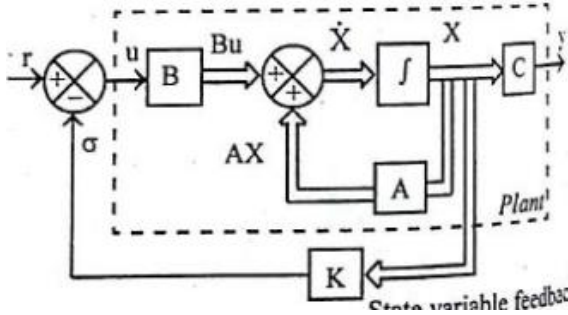


Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.Tech Degree 7th semester (S,FE) Exam April 2025 (2019 Scheme)			
Course Code: EET401			
Course Name: ADVANCED CONTROL SYSTEMS			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		Minimum set of variables, variables associated with energy storage elements, examples – 3 marks	(3)
2		$\dot{x} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u$, $y = \begin{bmatrix} 40 \\ 3 & -15 & 5 \end{bmatrix} x$	(3)
3		1+1+1	(3)
4		Steps-2 final-1	(3)
5		Modified state model-1 block diagram-2	(3)
			
6		Controllability-1.5, observability-1.5	(3)
7		Stability and instability Conditions (1.5+1.5)	(3)
8		2 types- 2, examples- 1	(3)
9		1+1+1	(3)
10		Indefinite ..Steps-1 answer -2	(3)
PART B			
Answer one full question from each module, each carries 14 marks.			
		Module I	
11	a)	$\dot{x} = \begin{bmatrix} -10 & -2 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} 6 \\ 0 \end{bmatrix} u$	(9)

	b)	- characteristic equation of this matrix is given by $\lambda^2 - 3\lambda + 2$ Eigenvalues of matrix A are $\lambda_1 = -1, \lambda_2 = -2$ The corresponding eigenvectors of the eigenvalues λ_1 and λ_2 are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ respectively - The modal matrix formed by these eigenvectors is given by $\mathbf{P} = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}$ - the diagonal matrix $\mathbf{\Lambda}$ is obtained by the transformation $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ $\mathbf{\Lambda} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$	(5)
		OR	
12	a)	$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{K_1}{M_1} & \frac{K_1}{M_1} & -\frac{B_1}{M_1} & \frac{B_1}{M_1} \\ \frac{K_1}{M_2} & -\frac{(K_1 + K_2)}{M_2} & \frac{B_1}{M_2} & -\frac{(B_1 + B_2)}{M_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M_1} \\ 0 \end{bmatrix} [u]$ $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$	(7)
	b)	So, $\dot{x}_1 = -x_1 - u$ $\dot{x}_2 = -(x_2 + x_1) = -(x_2 - x_1 - u)$ $\dot{x}_2 = x_1 - x_2 + u$ $y = \dot{x}_2$ $y = x_1 - x_2 + u$ $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$ $\dot{x} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} x + \begin{bmatrix} -1 \\ 1 \end{bmatrix} u$	(7)
		Module II	
13	a)	STM = $\begin{bmatrix} 3e^{-2t} - 2e^{-3t} & e^{-2t} - e^{-3t} \\ -6e^{-2t} + 6e^{-3t} & -2e^{-2t} + 3e^{-3t} \end{bmatrix}$ (3marks) $y(t) = \frac{1}{6} - \frac{1}{2}e^{-2t} + \frac{1}{3}e^{-3t}$ (4marks)	(8)

	b)	$f(\lambda) = e^{\lambda t} = \alpha_0 + \alpha_1 \lambda$ $\therefore f(-1) = e^{-t} = \alpha_0 - \alpha_1$ $\frac{d}{d\lambda} f(\lambda)_{\lambda=-1} = \frac{d}{d\lambda} e^{\lambda t}_{\lambda=-1} = \frac{d}{d\lambda} (\alpha_0 + \alpha_1 \lambda)$ $\Rightarrow te^{-t} = \alpha_1$ - Substituting the value of α_1 in equation(4) we get $\alpha_0 = (1+t)e^{-t}$ <u>Step-3:</u> - The required result is $f(\mathbf{A}) = e^{\mathbf{A}t} = \alpha_0 \mathbf{I} + \alpha_1 \mathbf{A}$ $= \begin{bmatrix} (1+t)e^{-t} & te^{-t} \\ -te^{-t} & (1-t)e^{-t} \end{bmatrix}$	(6)
		OR	
14	a)	$TF = \frac{1}{s^3 + s^2 + 1}$	(8)
	b)	Theorem -3 marks, Example -3 marks	(6)
		Module III	
15	a)	$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$ $Q_c = [B \quad AB \quad A^2B]$ $\therefore Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & a_3 \\ 1 & a_3 & a_2 + a_3^2 \end{bmatrix}$ $\therefore Q_c \neq 0$ $\therefore$ The system is controllable.	(7)
	b)	Controllability check-2, $u = -[29.6 \quad 3.6]x - 5$ marks	(7)
		OR	
16	a)	$\dot{x} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u, \quad y = [1 \quad 0 \quad 0]x$ Controllable and observable	(8)

		State model (2 marks), controllability(3marks), Observability(3marks)	
	b)	$K_e = \begin{bmatrix} 120.6 \\ 20 \end{bmatrix}$	(6)
		Module IV	
17	a)	Polar plot- 4 marks $-\frac{1}{K_N}$ plot- 3 Marks Frequency- 1.41 rad/sec Amplitude – 2.14 (1.5 each)	(10)
	b)	Output contains only the fundamental component, non-linearity have symmetry, and the linear part of the systems have low pass filter characteristics	(4)
		OR	
18	a)	Schematics- 2, derivation steps -4 , final equation- 2 marks	(8)
	b)	2+2+2	(6)
		Module V	
19	a)	Stable and unstable limit cycles- 2marks each	(4)
	b)	$K > 0$ Steps -5 final answer - 5	(10)
		OR	
20	a)	Singular points (0,0) and (2,0) State model-2 marks singular points -2 each $\begin{aligned} x_1 &= x \\ x_2 &= \dot{x}, \end{aligned}$ the corresponding first-order system is given by $\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -5x_1 - 4x_2 + 2.5x_1^2 \end{bmatrix}.$ • Identify the singular points We then compute the singular points for such a system: they are defined by the equations: $x_2 = 0$, $-5x_1 + 2.5x_1^2 = 0$, thus leading to the two singular points (0, 0) and (2, 0).	(6)
	b)	Asymptotically stable $\dot{V}(x) = -2x_1^4 - 2x_2^2$	(8)
